# AUTOMATED BONE FRACTURE DETECTION & LOCALIZATION

*A 3-Stage Deep Learning Pipeline on FracAtlas*

Computer Vision · Final Project Defense

Saint Joseph University · Faculty of Engineering · Spring 2026

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# The Clinical Problem

*Why bone fracture detection on X-rays matters*

**178 M**

*new fracture cases globally each year*

(World Health Organization, 2023)

X-ray remains the first-line imaging modality (low cost, wide availability, fast acquisition)

### Why is this hard?

- Hairline cracks, displaced fragments, and subtle cortical disruptions are routinely missed by junior clinicians under emergency time pressure.

### Our two-part clinical question

1. Is a fracture present in this X-ray? → classification
2. If so, where exactly on the bone? → localization

# Our Approach — A 3-Stage Pipeline

*Classify → Localize → Explain*

## 1 Classification

`EfficientNet-B3`

4-class output:
hand/leg × fractured/normal

**86.6% accuracy**

## 2 Localization

`YOLOv8s / Faster R-CNN`

Bounding boxes around
the suspected fracture

**0.62 F1 (FRCNN)**

## 3 Explainability

`Grad-CAM`

Spatial saliency map
over the classifier's input

**visual rationale**

**Key design choice:** Stage 2 only runs when Stage 1 flags a fracture

# The Dataset Journey

*It took us four attempts to find the right corpus*

## v1

**Binary + multi merge**

*Result:*

**96% train / 50% val**

*Why:*

Binary dataset (fractured / not fractured) with multi-class one (10 fracture types) for a 2-stage approach.
Failed: mixed modalities, brightness bias (μ=51 vs 84–110)

## v2

**Same data, fewer classes**

*Result:*

**Val ceiling 47–50%**

*Why:*

Merged the 11 classes down to 6.
The val ceiling barely moved: data was the bottleneck, not the model

## v3

**Single-source switch (pkdarabi)**

*Result:*

**Val 32%**

*Why:*

~84 imgs/class for 10 classes — fundamentally insufficient

## v4

**FracAtlas (Nature)**

*Result:*

**Test 86.6% ✓**

*Why:*

Peer-reviewed, single-source, X-ray-only, with bounding-box annotations

***Lesson:*** *no architecture, learning rate, or augmentation strategy can compensate for inconsistent data.*

# FracAtlas — Our Final Dataset

*Islam et al. (2023), Scientific Data, Nature*

## 4,083 X-rays

*From 3 hospitals, annotated by 2 radiologists + 1 orthopedic surgeon*

### Per-image labels

- Body region (hand, leg, hip, shoulder)
- Fracture status (fractured / not)
- Bounding boxes (COCO, VOC, YOLO, VGG)
- Projection view, instance count, hardware


### From 8 classes to 4

| Class | Train | Status |
|---|---|---|
| hand_fractured | 284 | kept |
| hand_normal | 676 | kept |
| leg_fractured | 159 | kept |
| leg_normal | 1,434 | kept |
| hip_fractured | 7 | EXCLUDED |
| shoulder_fractured | 7 | EXCLUDED |

**Why excluded?** *Only 7 fractured training examples each → 204× imbalance on the 8-class formulation*

# Stage 1 — EfficientNet-B3 Classifier

*Two-phase fine-tuning with class-weighted loss*

## Why EfficientNet-B3

- 10.7 M parameters
- 300 × 300 input — preserves the spatial detail of hairline fractures
- ImageNet-pretrained — leverages transfer learning
- Replace 1000-class head with Linear(4) + Dropout(0.4)

**No ColorJitter**

*Brightness jitter would corrupt the diagnostic signal.*

## Two-Phase Fine-Tuning

**Phase 1** *(epochs 1–15)*

`Backbone FROZEN · LR_head = 3e-3`

**Phase 2** *(epoch 16+, early stop @ patience 12)*

`Backbone UNFROZEN · LR_backbone = 1e-5`

`LR_head = 1e-4`

# Threshold Tuning — Free +7 pp F1

*A post-hoc calibration that adds zero training cost*

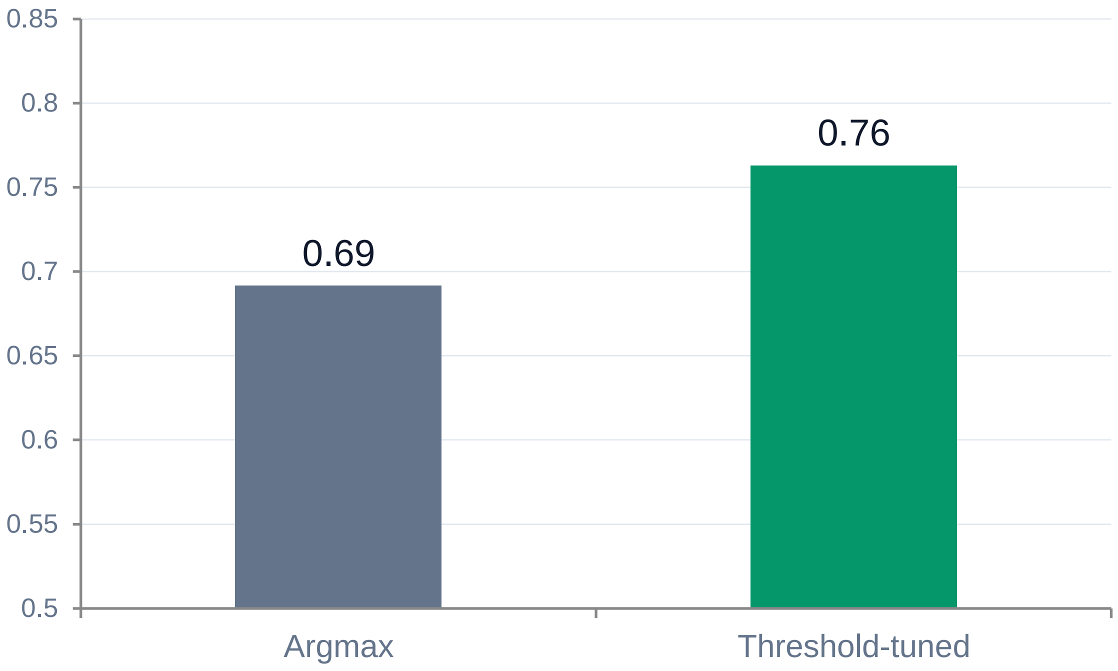
### The intuition

- Argmax assumes balanced base rates
- Our classes are imbalanced (9× between leg_normal and leg_fractured)
- Optimal per-class boundary is no longer at p = 0.5

#### Our procedure

Sweep threshold t ∈ [0.10, 0.90] per class on the validation set; keep the t that maximizes per-class F1. Persist in pkl, reuse at test time.

**Validation Macro F1 — before vs after**



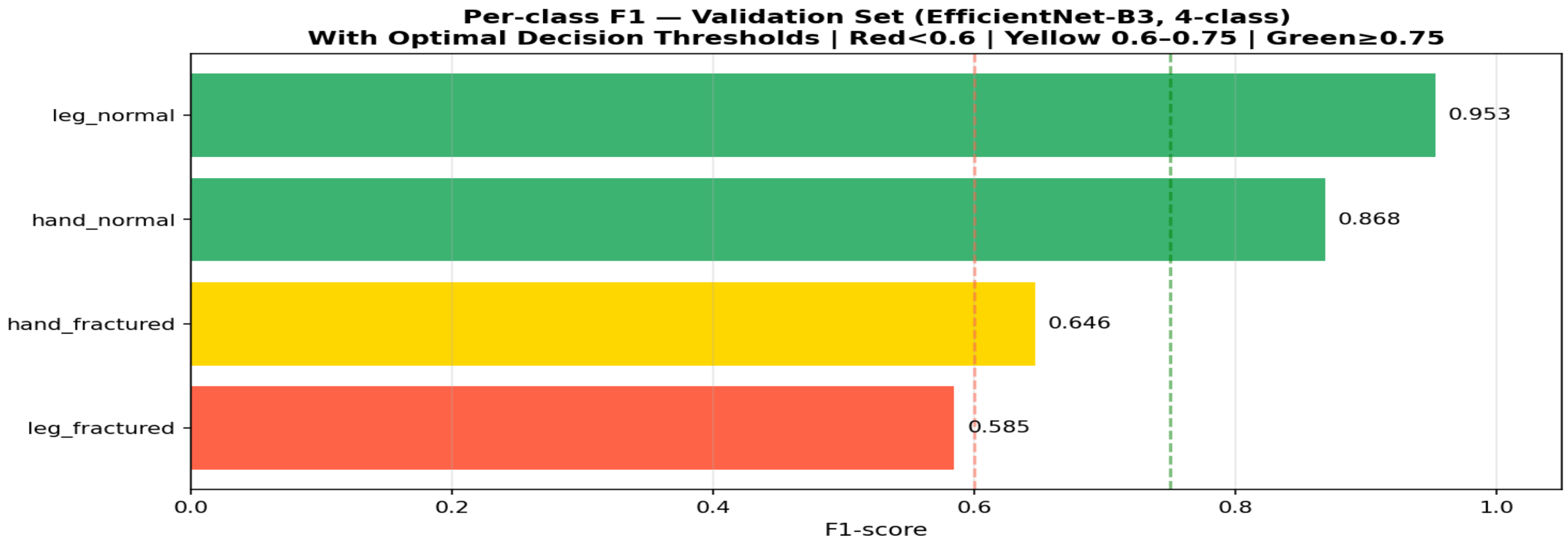

**Final thresholds:** `[0.50, 0.32, 0.50, 0.10]`

# Stage 1 Results — Test Set (344 images)

*Held out from all training and validation*

| 86.6% | 0.7625 | 0.95 | 0.57 |
|---|---|---|---|
| *Accuracy* | *Macro F1* | *leg_normal F1* | *leg_fractured F1* |

## Per-class F1 on the test set




**Clinical reading:** *Recall on leg_fractured (0.59) matters more than precision (0.54) — missing a fracture is costlier than a false alarm.*

# Stage 2 — Why Two Detectors?

*Single-stage vs two-stage: a fair side-by-side comparison*

## YOLOv8s

*1-stage anchor-free detector*

- 11.1 M parameters
- 640 × 640 input
- COCO-pretrained init
- Conservative augmentation (hsv_h = 0 for grayscale)
- No mixup (blurs fracture line orientation)

**Optimized for:** *speed & precision*

## Faster R-CNN

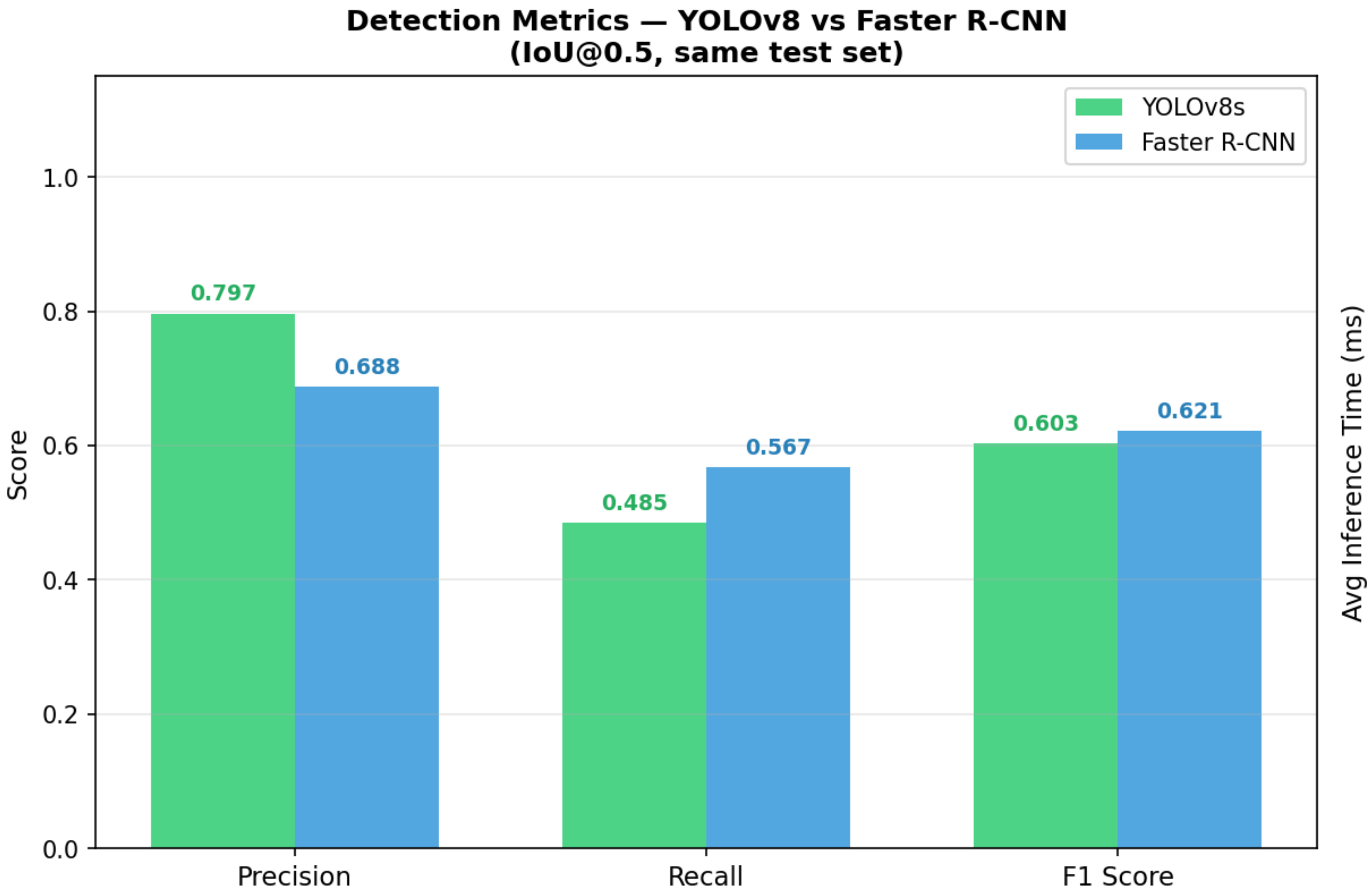
*2-stage ResNet-50 FPN detector*

- ~41.8 M parameters
- Region Proposal Network → RoI pooling → classification head
- ImageNet-pretrained backbone
- Two-class output (background / fracture)
- Confidence threshold 0.50

**Optimized for:** *recall & F1*

# Detection Comparison — Hand+Leg Filtered Test Set

*IoU 0.5 hard matching, identical scope*




### YOLOv8s wins on...

Precision (+0.08)
Speed: 32 ms / image (4× faster)

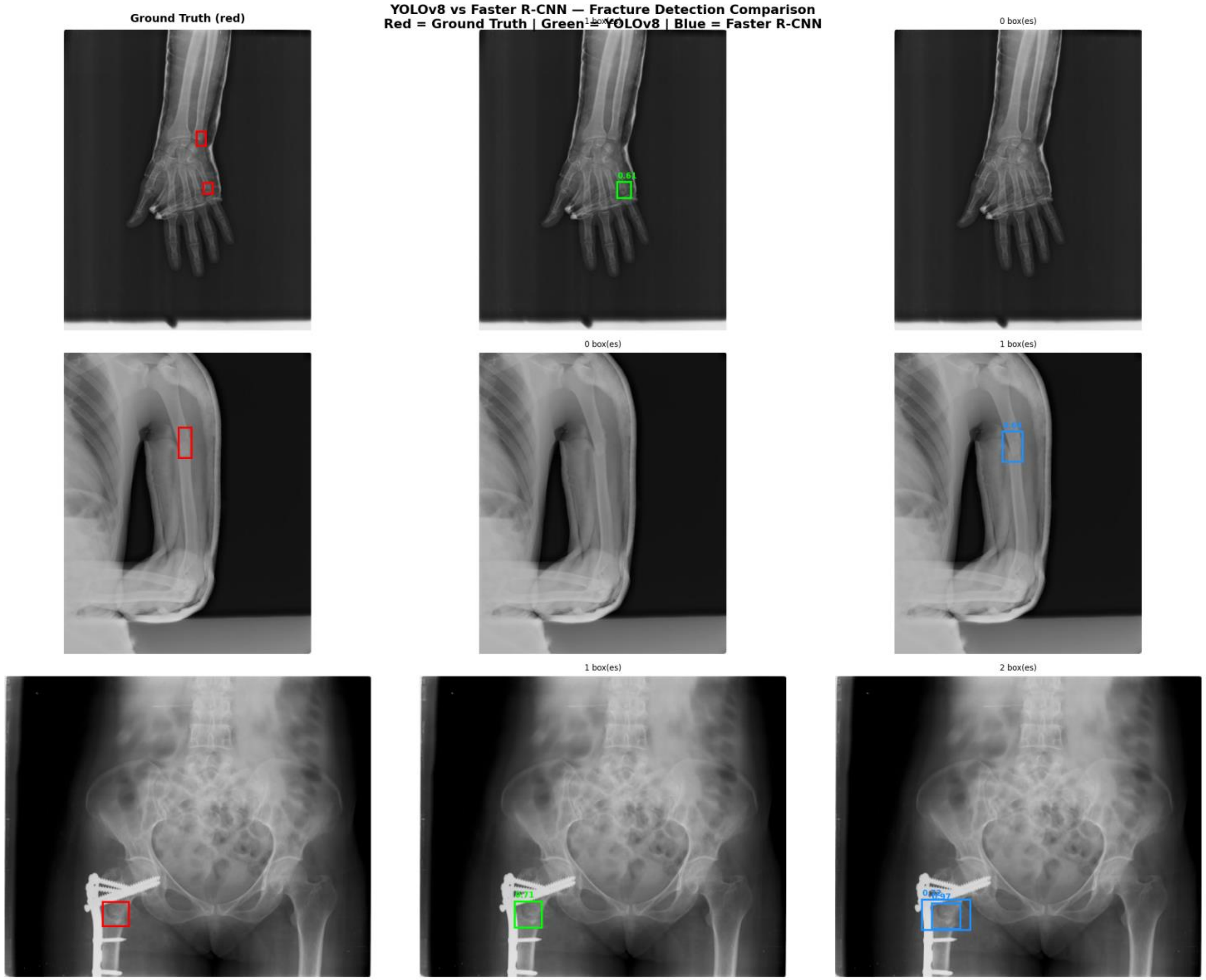
### Faster R-CNN wins on...

Recall (+0.09)
F1 (+0.03)
Latency: 126 ms / image

### Clinical verdict

Faster R-CNN preferred — missing a fracture is more costly than a false alarm.

# Detection Comparison — Hand+Leg Filtered Test Set



# Stage 3 — Grad-CAM Explainability

*Visual rationale for every classifier decision*

### Why explainability matters

- Lets clinicians review the model's reasoning instead of trusting a raw score
- Catches spurious correlations (e.g. background features driving the prediction)
- Required for clinical decision-support tools

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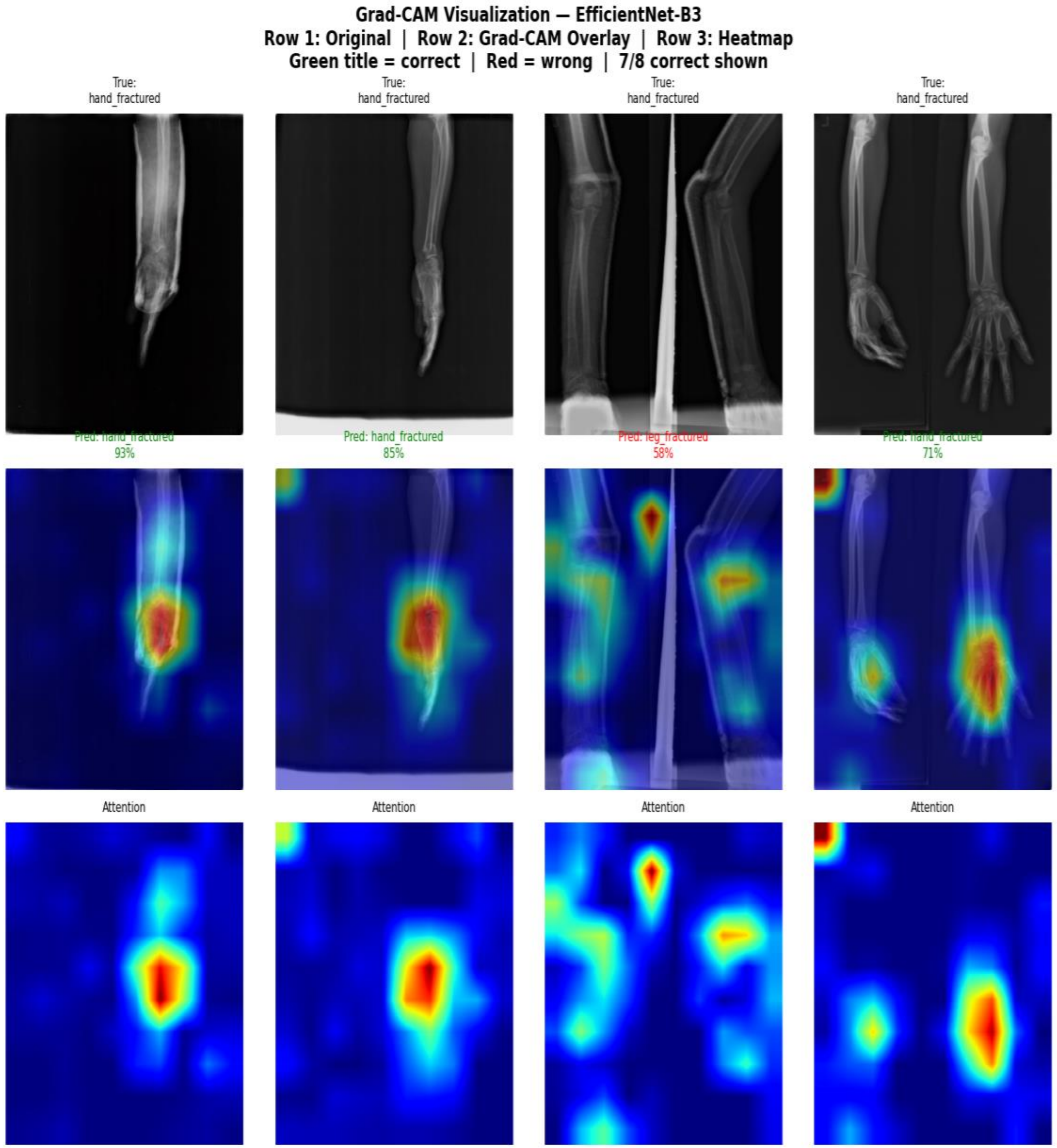
### Implementation

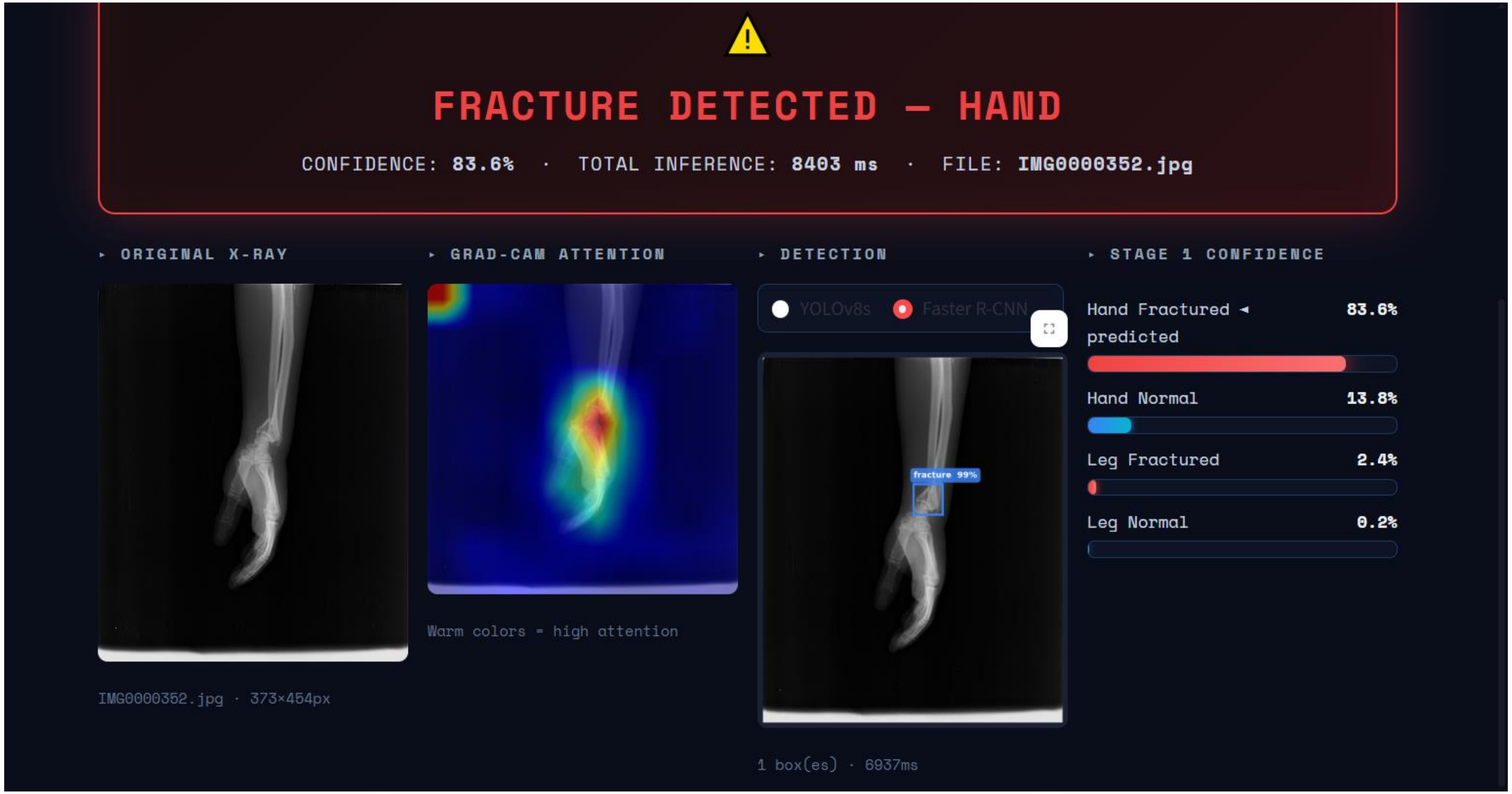
*Target: features[-1] — the last MBConv block of EfficientNet-B3. Highest-level features that still preserve spatial resolution.*




*Warm colors = high attention*

# Deployed System — Interactive Web Application

*Streamlit-based real-time inference on uploaded X-rays*




**Per-stage panels**

Original X-ray, Grad-CAM heatmap, detector bounding boxes, and Stage-1 confidence bars — all in one view.

**Detector toggle**

Switch between YOLOv8 and Faster R-CNN at inference time, reusing the cached classifier output.

**Calibrated decisions**

Stage-1 thresholds [0.50, 0.32, 0.50, 0.10] applied automatically. Detector skipped on normal X-rays.

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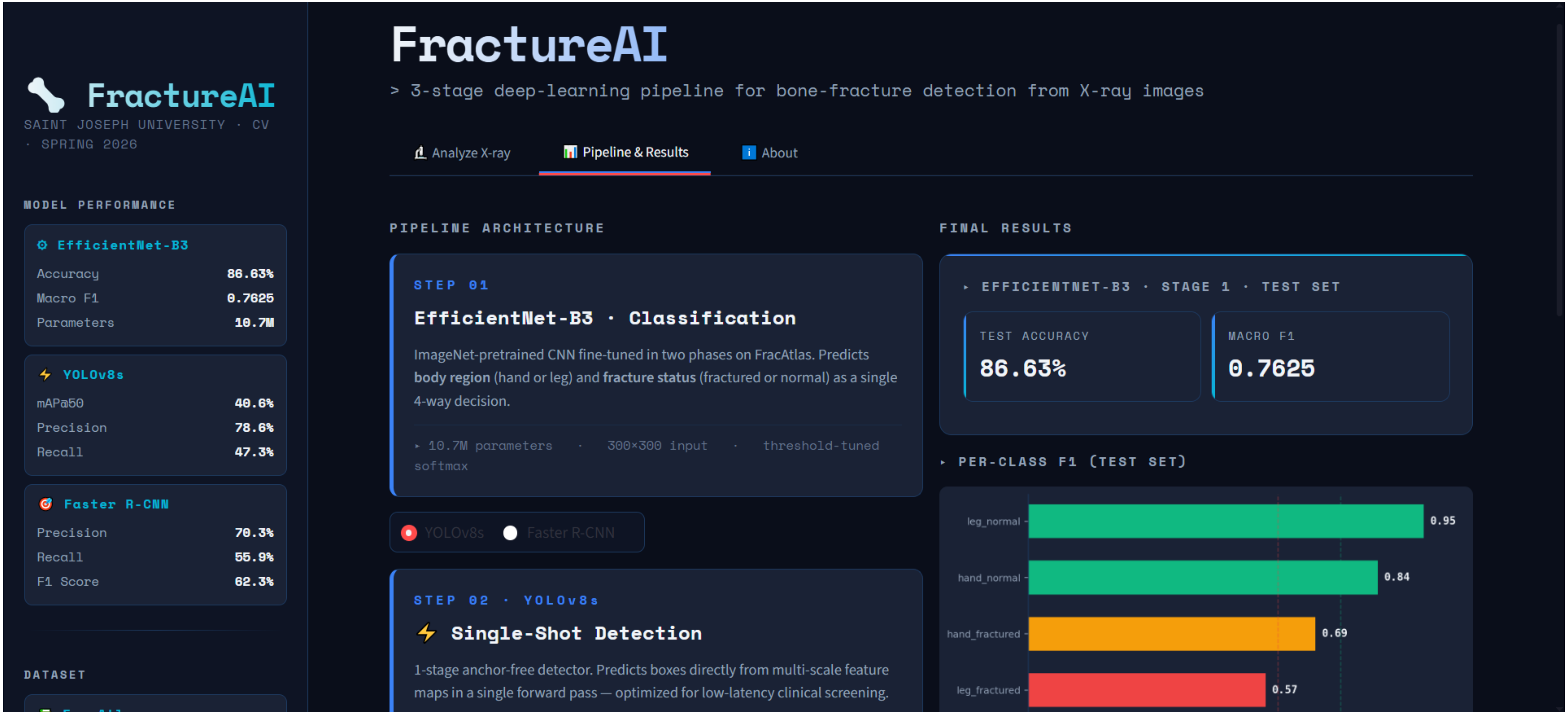

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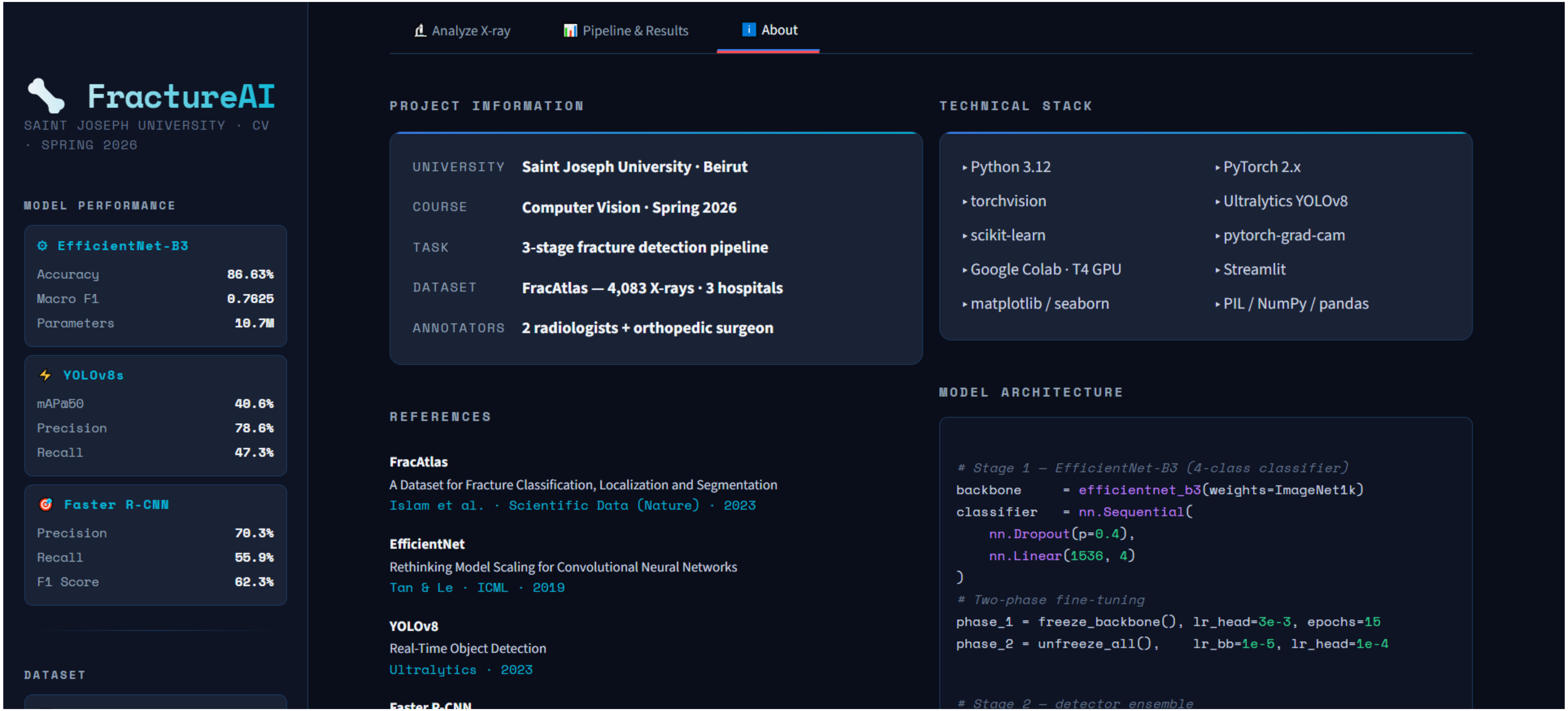

# Limitations & Future Work

*What we'd do with more time*

## Honest limitations

- Hand and leg only — hip/shoulder excluded due to scarcity
- Only 3 hospitals — limited inference about scanner robustness
- No external out-of-distribution evaluation


## Future work (priority order)

1. Hierarchical router: hand/leg vs hip/shoulder so excluded regions are not silently dropped
2. Evaluate on an external dataset to characterize domain transfer
3. Port the pipeline to ONNX / TensorRT for embedded deployment (bonus rubric points)

# Conclusion

*Three takeaways from this project*

1

## Dataset quality dominates

Three failed dataset versions taught us that no architecture or LR schedule compensates for inconsistent data. FracAtlas was the turning point.

2

## Free wins exist

Post-hoc threshold tuning added ~7 percentage points of Macro F1 with zero training cost. Always calibrate.

3

## Fair comparisons matter

Side-by-side YOLOv8 vs Faster R-CNN on a scope-matched test set produced a clinically actionable trade-off — neither model is universally better.

**Thank you**